

PROJECT APPRAISAL

Module 4

**Financial Discount Rates in Project
Appraisal**



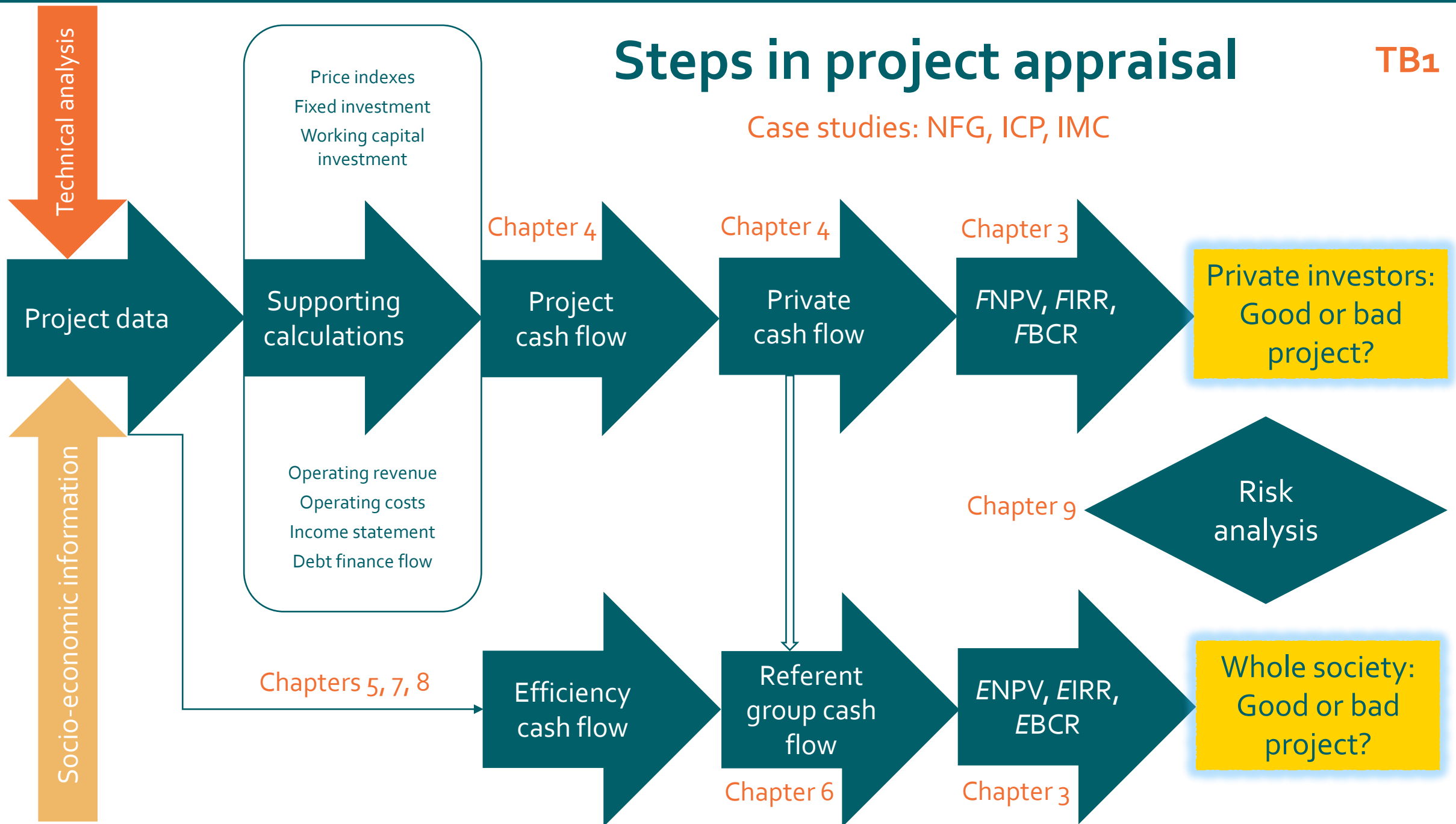
Outline – Module 4

- Interest tax shield
- Financial cash flow re-considerations
- Financial discount rates: An introduction
- Financial discount rates: No income tax
- Financial discount rates: Impact of income tax
- Practical implications
- An illustrative example

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Tax shield

With and without interest expense matter



Interest tax shield

- One of the main objectives of the project's owners is to reduce their **tax liability** as much as possible.
- Interest tax shield refers to the reduced income taxes brought about by deductions to taxable income from a project's interest expense.
- Therefore, interest tax shields encourage equity holders to finance their projects with debt.
- Interest tax shield can be defined as:

$$TS_t = D_t \cdot d \cdot t_i$$

where D_t : the beginning debt at year t , d : cost of debt (i.e. interest rate), and t_i : income tax rate.

Interest tax shield

	A	B	C	D	E	F	G	H	I	J	K
5	TOTAL INVESTMENT POINT OF VIEW (TIPV)						ALL EQUITY POINT OF VIEW (AEPV)				
6	Revenues	1000					Revenues	1000			
7	Operating costs	500					Operating costs	500			
8	Depreciation	333					Depreciation	333			
12	Investment costs	1000					Investment costs	1000			
13	Debt financing	100%					Debt financing	0%			
14	Interest rate	10%	Income tax rate				Interest rate				
15	Interest expense	IPMT	20%				Interest expense	0			
16											
17	<u>Income statement</u>						<u>Income statement</u>				
18	Year	0	1	2	3		Year	0	1	2	3
19	Revenues		1000	1000	1000		Revenues		1000	1000	1000
20	Operating costs		500	500	500		Operating costs		500	500	500
21	EBITDA		500	500	500		EBITDA		500	500	500
22	Depreciation		333	333	333		Depreciation		333	333	333
23	EBIT		167	167	167		EBIT		167	167	167
24	Interest expense		100	70	37		Interest expense	0	0	0	0
25	EBT		67	97	130		EBT		167	167	167
26	Income tax		13	19	26		Income tax		33	33	33
27	Interest tax shield		20	14	7		Interest tax shield	0	0	0	0

Interest tax shield

	A	B	C	D	E	F	G	H	I	J	K
27	Interest tax shield		20	14	7		Interest tax shield		0	0	0
28											
29	Cash flow statement						Cash flow statement				
30	Year	0	1	2	3		Year	0	1	2	3
31	Cash receipts		1000	1000	1000		Cash receipts		1000	1000	1000
32	Revenues		1000	1000	1000		Revenues		1000	1000	1000
33	Cash expenditures	1000	500	500	500		Cash expenditures	1000	500	500	500
34	Investment cost	1000					Investment cost	1000			
35	Operating costs		500	500	500		Operating costs		500	500	500
36	Market net cash flow	-1000	500	500	500		Market net cash flow	-1000	500	500	500
37	Income tax		13	19	26		Income tax		33	33	33
38	Net cash flow (TIPV)	-1000	487	481	474		Net cash flow (AEPV)	-1000	467	467	467
39	$NCF_{TIPV} - NCF_{AEPV}$		20	14	7						
40											
41	Discount rates	5%	10%	15%	20%		Discount rates	5%	10%	15%	20%
42	NPV_{TIPV}	309	196	98	14		NPV_{AEPV}	271	161	66	(17)
43	PV_{TS}	38	35	33	31		$NPV_{AEPV} + PV_{TS}$	309	196	98	14

REMARKS: 1) NCF_{TIPV} is larger than NCF_{AEPV} ; 2) The difference between NCF_{TIPV} and NCF_{AEPV} equals the interest tax shield; 3) NPV_{TIPV} is higher than NPV_{AEPV} ; 4) The sum of NPV_{AEPV} and PV_{TS} equals NPV_{TIPV} .

Interest tax shield

	A	B	C	D	E	F	G	H	I	J	K
27	Interest tax shield		30	21	11		Interest tax shield		0	0	0
28											
29	Cash flow statement						Cash flow statement				
30	Year	0	1	2	3		Year	0	1	2	3
31	Cash receipts		1000	1000	1000		Cash receipts		1000	1000	1000
32	Revenues		1000	1000	1000		Revenues		1000	1000	1000
33	Cash expenditures	1000	500	500	500		Cash expenditures	1000	500	500	500
34	Investment cost	1000					Investment cost	1000			
35	Operating costs		500	500	500		Operating costs		500	500	500
36	Market net cash flow	-1000	500	500	500		Market net cash flow	-1000	500	500	500
37	Income tax		3	12	22		Income tax		33	33	33
38	Net cash flow (TIPV)	-1000	497	488	478		Net cash flow (AEPV)	-1000	467	467	467
39	$NCF_{TIPV} - NCF_{AEPV}$		30	21	11						
40											
41	Discount rates	5%	10%	15%	20%		Discount rates	5%	10%	15%	20%
42	NPV_{TIPV}	329	214	115	29		NPV_{AEPV}	271	161	66	(17)
43	PV_{TS}	58	54	50	46		$NPV_{AEPV} + PV_{TS}$	329	214	115	29

REMARKS: 1) Higher interest rate, i.e. **15%** now, leads to higher interest tax shield; 2) This then results in higher NPV_{TIPV} ; 3) But NPV_{AEPV} remains unchanged. **Note #1: NCF_{TIPV} is the tax shield inclusive NCF.**

Interest tax shield

	A	B	C	D	E	F	G	H	I	J	K
27	Interest tax shield		10	7	4		Interest tax shield		0	0	0
28											
29	Cash flow statement						Cash flow statement				
30	Year	0	1	2	3		Year	0	1	2	3
31	Cash receipts		1000	1000	1000		Cash receipts		1000	1000	1000
32	Revenues		1000	1000	1000		Revenues		1000	1000	1000
33	Cash expenditures	1000	500	500	500		Cash expenditures	1000	500	500	500
34	Investment cost	1000					Investment cost	1000			
35	Operating costs		500	500	500		Operating costs		500	500	500
36	Market net cash flow	-1000	500	500	500		Market net cash flow	-1000	500	500	500
37	Income tax		23	26	30		Income tax		33	33	33
38	Net cash flow (TIPV)	-1000	477	474	470		Net cash flow (AEPV)	-1000	467	467	467
39	$NCF_{TIPV} - NCF_{AEPV}$		10	7	4						
40											
41	Discount rates	5%	10%	15%	20%		Discount rates	5%	10%	15%	20%
42	NPV_{TIPV}	290	178	82	(2)		NPV_{AEPV}	271	161	66	(17)
43	PV_{TS}	19	18	16	15		$NPV_{AEPV} + PV_{TS}$	290	178	82	(2)

REMARKS: 1) Lower debt financing, i.e. **50%** now, leads to lower interest tax shield; 2) This then results in lower NPV_{TIPV} ; 3) But NPV_{AEPV} remains unchanged. **Note #2: NCF_{AEPV} is the tax shield exclusive NCF.**

Financial cash flows

Different points of view re-consideration



Cash flow statement (TIPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) ΔAR					
d) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) VAT on purchases					
d) ΔAP					
e) ΔCB					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2c)					
b) Income tax, including interest tax shield					
5. Market net cash flow (TIPV)	= (3) – (4)				

CFS with the
tax shield

Cash flow statement

(TIPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) Changes in working capital					
d) VAT on purchases					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2d)					
b) Income tax, including interest tax shield					
5. Market net cash flow (TIPV)	= (3) – (4)				

CFS with the
tax shield

Cash flow statement

(AEPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) ΔAR					
d) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) VAT on purchases					
d) ΔAP					
e) ΔCB					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2c)					
b) Income tax, excluding interest tax shield					
5. Market net cash flow (AEPV)	= (3) – (4)				

CFS without
the tax shield

It's also called **Total Investment Cashflow** without tax shield.

Cash flow statement

(AEPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) Changes in working capital					
d) VAT on purchases					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2d)					
b) Income tax, excluding interest tax shield					
5. Market net cash flow (TIPV)	= (3) – (4)				

CFS without
the tax shield

It's also called **Total Investment Cashflow** without tax shield.

Cash flow statement

(EPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) ΔAR					
d) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) VAT on purchases					
d) ΔAP					
e) ΔCB					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2c)					
b) Income tax, including interest tax shield					
5. Debt finance flow					
6. Private net cash flow (EPV)	= (3) – (4) – (5)				

CFS with the
tax shield

$$NCF_{EPV} = NCF_{TIPV} + \text{Debt finance flow}$$

Cash flow statement

(EPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) Changes in working capital					
d) VAT on purchases					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2d)					
b) Income tax, including interest tax shield					
5. Debt finance flow					
6. Market net cash flow (TIPV)	= (3) – (4) – (5)				

CFS with the
tax shield

$$NCF_{EPV} = NCF_{TIPV} + \text{Debt finance flow}$$

Financial discount rates

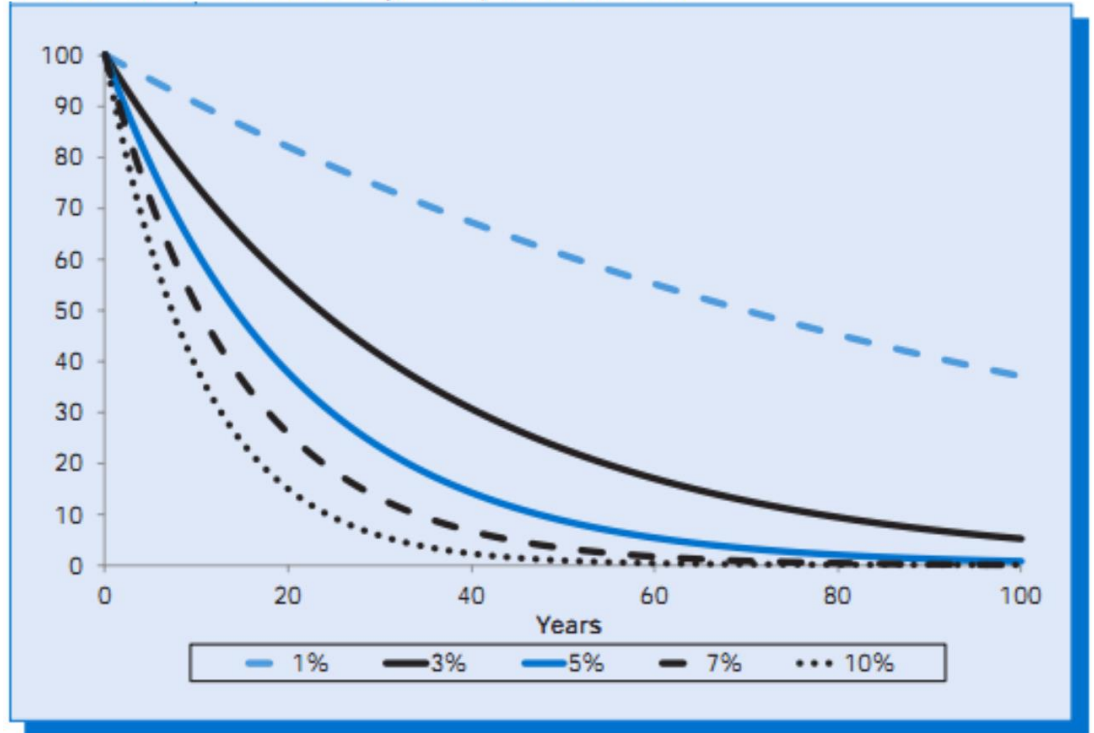
An introduction



Financial discount rates

- The higher discount rates dramatically reduce the relevance of impacts that occur just a few decades in the future.
- With a 5 percent discount rate it would not be worth spending even \$1 now to avoid a damage of \$100 that occurs 100 years in the future.
- With a 10 percent discount rate it would not be worth spending even \$1 now to receive a benefit of \$100 that occurs 50 years in the future.
- The choice of a discount rate is an important decision in any project appraisal.

Present Value of a \$100 Impact, by Discount Rate



Present Value of a \$100 Impact, by Discount Rate

Years into the Future	Discount Rate				
	1%	3%	5%	7%	10%
0	\$100.00	\$100.00	\$100.00	\$100.00	\$100.00
10	\$90.53	\$74.41	\$61.39	\$50.83	\$38.55
20	\$81.95	\$55.37	\$37.69	\$25.84	\$14.86
30	\$74.19	\$41.20	\$23.14	\$13.14	\$5.73
50	\$60.80	\$22.81	\$8.72	\$3.39	\$0.85
100	\$36.97	\$5.20	\$0.76	\$0.12	\$0.01

A high discount rate will highly favor the present over the future, while a low discount rate will give more weight to future costs or benefits.

Financial discount rates

- The financial discount rate is the **opportunity cost of funds** that are invested in the project. It depends upon the viewpoints of analysis.
 - When a project **without debt finance** is being appraised from *the point of view of total capital (or all equity holders)*, the relevant cost of funds is the unlevered return on equity that is being earned in its best alternative use.
 - When a project **with debt finance** is being appraised from *the point of view of total capital*, the relevant cost of funds is the average return on capital that is being earned in its best alternative use.
 - When a project **with debt finance** is being appraised from *the point of view of the equity capital*, the relevant cost of funds is the levered return on equity that is being earned in its best alternative use.

Financial discount rates

The **opportunity cost of funds** that are invested in the project can be classified as follows:

Source of funds	Cash flow statement	Opportunity cost	Symbol
1) Debt capital		Cost of debt	d
2) Equity capital	Debt financing	Cost of equity	e
3) Equity capital	No-debt financing	Cost of equity	ρ
4) Debt & equity	Debt financing, tax shield	Cost of capital	$wacc_1$
5) Debt & equity	Debt financing, no tax shield	Cost of capital	$wacc_2$

Financial discount rates

Debt
financing,
no income
tax

$$wacc_1 = \%Dd + \%Ee$$

$$\%D = \frac{D}{(D+E)}, \%E = \frac{E}{(D+E)}$$

$$e = \rho + (\rho - d) \frac{D}{E}$$

$$wacc_2 = \%Dd(1 - t) + \%Ee$$

$$e = \rho + (\rho - d) \frac{D}{E}$$

t: income tax rate

Debt
financing,
income
tax

Financial discount rates

No income tax



Financial discount rates: No income tax

Year	0	1
Revenue	0	1200
Investment	-1000	0
NCF_{TIPV}	-1000	1200

Suppose there is **no income tax** ($t = 0\%$).

Year	0	1
Revenue	0	1200
Investment	-1000	0
NCF_{AEPV}	-1000	1200

$$r_{TIPV} = \frac{1200 - 1000}{1000} = 20\%$$

No debt financing

$$r_{AEPV} = \frac{1200 - 1000}{1000} = 20\%$$

D = 0%

Financial discount rates: No income tax

Suppose, to finance the project, we borrow 20% of the investment cost at an interest rate of 8% per annum. Equity accounts 80% of investment. Thus, D/E is 0.25.

Year	0	1
NCF_{TIPV}	-1000	1200
Debt finance flow	200	-216
NCF_{EPV}	-800	984

$$\rho = \frac{1200 - 1000}{1000} = 20\%$$

$$D = 20\%$$

$$d = 8\%$$

$$e = \frac{984 - 800}{800} = 23\%$$

$$D = 20\%$$

Financial discount rates: No income tax

Suppose, to finance the project, we borrow 40% of the investment cost at an interest rate of 8% per annum. Equity accounts 60% of investment. Thus, D/E is 0.667.

Year	0	1
NCF_{TIPV}	-1000	1200
Debt finance flow	400	-432
NCF_{EPV}	-600	768

$$\rho = \frac{1200 - 1000}{1000} = 20\%$$

$$D = 40\%$$

$$d = 8\%$$

$$e = \frac{768 - 600}{600} = 28\%$$

$$D = 40\%$$

Financial discount rates: No income tax

Suppose, to finance the project, we borrow 60% of the investment cost at an interest rate of 8% per annum. Equity accounts 40% of investment. Thus, **D/E is 1.5**.

Year	0	1
NCF_{TIPV}	-1000	1200
Debt finance flow	600	-648
NCF_{EPV}	-400	552

$$\rho = \frac{1200 - 1000}{1000} = 20\%$$

$$D = 60\%$$

$$d = 8\%$$

$$e = \frac{552 - 400}{400} = 38\%$$

$$D = 60\%$$

Financial discount rates: No income tax

Suppose, to finance the project, we borrow 80% of the investment cost at an interest rate of 8% per annum. Equity accounts 20% of investment. Thus, **D/E is 4**.

Year	0	1
NCF_{TIPV}	-1000	1200
Debt finance flow	800	-864
NCF_{EPV}	-200	336

$$\rho = \frac{1200 - 1000}{1000} = 20\%$$

$$D = 80\%$$

$$d = 8\%$$

$$e = \frac{336 - 200}{200} = 68\%$$

$$D = 80\%$$

Financial discount rates: No income tax

$$\rho = 20\% \quad d = 8\% \quad e = \rho + (\rho - d) \frac{D}{E} \quad wacc_1 = \%Dd + \%Ee$$

Case	%D	%E	D/E	e	wacc ₁
1	0%	100%	0.00	20%	20%
2	20%	80%	0.25	23%	20%
3	40%	60%	0.67	28%	20%
4	60%	40%	1.50	38%	20%
5	80%	20%	4.00	68%	20%

Financial discount rates: No income tax

$$\rho = 20\% \quad d = 12\% \quad e = \rho + (\rho - d) \frac{D}{E} \quad wacc_1 = \%Dd + \%Ee$$

Case	%D	%E	D/E	e	wacc ₁
1	0%	100%	0.00	20.0%	20%
2	20%	80%	0.25	22.0%	20%
3	40%	60%	0.67	25.3%	20%
4	60%	40%	1.50	32.0%	20%
5	80%	20%	4.00	52.0%	20%

Financial discount rates: No income tax

$$\rho = 15\% \quad d = 8\% \quad e = \rho + (\rho - d) \frac{D}{E} \quad wacc_1 = \%Dd + \%Ee$$

Case	%D	%E	D/E	e	wacc ₁
1	0%	100%	0.00	15.0%	15%
2	20%	80%	0.25	16.8%	15%
3	40%	60%	0.67	19.7%	15%
4	60%	40%	1.50	25.5%	15%
5	80%	20%	4.00	43.0%	15%

Financial discount rates: No income tax

$$\rho = 15\% \quad d = 12\% \quad e = \rho + (\rho - d) \frac{D}{E} \quad wacc_1 = \%Dd + \%Ee$$

Case	%D	%E	D/E	e	wacc ₁
1	0%	100%	0.00	15.0%	15%
2	20%	80%	0.25	15.8%	15%
3	40%	60%	0.67	17.0%	15%
4	60%	40%	1.50	19.5%	15%
5	80%	20%	4.00	27.0%	15%

Financial discount rates: No income tax

■ Remarks:

- ρ and $wacc_1$ are always identical.
 - As D/E increases, e increases; and e is greater than ρ when the project is financed by debt.
 - As d increases, e decreases.
 - As ρ increases, e increases.
- With debt financing, the risk is higher for the equity holder and thus the return must be higher to compensate for the higher risk.

Financial discount rates: No income tax

- In the absence of taxes, the value of the levered project (V^L) should be equal to the value of the unlevered project (V^{UL}).
 - Value of the project at year t is the sum of discounted cash flow at *correct discount rate* from year $t+1$ to the end of the project.
- In turn, the value of the levered project is equal to the value of the equity (E^L) and the value of the debt (D): $V^L = E^L + D$.
- In this case, the correct discount rate for NCF_{TIPV} (i.e. $wacc_1$) is equal to the required return on all-equity financing, namely ρ .

Financial discount rates: No income tax

We have the following formulas:

$$PV[NCF_{TIPV}]_{@wacc1} = PV[NCF_{AEPV}]_{@p}$$

This is
always
zero

$$PV[NCF_{TIPV}]_{@wacc1} = PV[NCF_{EPV}]_{@e} + PV[NCF_{Debt}]_{@d} = \\ PV[NCF_{AEPV}]_{@p}$$

$$NPV[NCF_{TIPV}]_{@wacc1} = NPV[NCF_{EPV}]_{@e} + NPV[NCF_{Debt}]_{@d} = \\ NPV[NCF_{AEPV}]_{@p}$$

So,
 NPV_{TIPV}
=
 NPV_{EPV}

Year	0	1
Revenue	0	1200
Investment	-1000	0
NCF_{TIPV}	-1000	1200

Year	0	1
Revenue	0	1200
Investment	-1000	0
NCF_{AEPV}	-1000	1200

Year	0	1
NCF_{TIPV}	-1000	1200
Debt finance flow	400	-432
NCF_{EPV}	-600	768

We've already known:

- $\rho = 20\%$ ($= r_{AEPV} = 20\%$)
- $d = 8\%$
- $e = 28\% = 20\% + (20\% - 8\%) * 0.667$
- $wacc_1 = 20\% = 40\% * 8\% + 60\% * 28\%$

Therefore,

- $NPV[NCF_{TIPV}]_{@wacc1 = 20\%} = 0$
- $NPV[NCF_{AEPV}]_{@ \rho = 20\%} = 0$
- $NPV[NCF_{EPV}]_{@e = 28\%} = 0$
- $NPV[NCF_{Debt}]_{@d = 8\%} = 0$

Year	0	1
Revenue	0	1250
Investment	-1000	0
NCF_{TIPV}	-1000	1250

Year	0	1
Revenue	0	1250
Investment	-1000	0
NCF_{AEPV}	-1000	1250

Year	0	1
NCF_{TIPV}	-1000	1250
Debt finance flow	400	-432
NCF_{EPV}	-600	818

In a same manner:

- $\rho = 20\%$ ($< r_{AEPV} = 25\%$)
- $d = 8\%$
- $e = 27.48\% = 20\% + (20\% - 8\%) * 0.623$
- $wacc_1 = 20\% = 38.4\% * 8\% + 61.6\% * 27.48\%$

Therefore,

- $NPV[NCF_{TIPV}]_{@wacc1 = 20\%} = 41.67$
- $NPV[NCF_{AEPV}]_{@p = 20\%} = 41.67$
- $NPV[NCF_{EPV}]_{@e = 27.48\%} = 41.67$
- $NPV[NCF_{Debt}]_{@d = 8\%} = 0$

Financial discount rates: No income tax

	A	B	C	D	E	F	G	H	I	J	K
5	TOTAL INVESTMENT POINT OF VIEW (TIPV)						ALL EQUITY POINT OF VIEW (AEPV)				
6	Revenues	1000					Revenues		1000		
7	Operating costs	500					Operating costs		500		
8	Depreciation	333					Depreciation		333		
12	Investment costs	1000	ρ 20%				Investment costs		1000		
13	Debt financing	40%					Debt financing		0%		
14	Interest rate	8%	Income tax rate				Interest rate				
15	Interest expense	IPMT					Interest expense		0		
16											
17	<u>Income statement</u>						<u>Income statement</u>				
18	Year	0	1	2	3		Year	0	1	2	3
19	Revenues		1000	1000	1000		Revenues		1000	1000	1000
20	Operating costs		500	500	500		Operating costs		500	500	500
21	EBITDA		500	500	500		EBITDA		500	500	500
22	Depreciation		333	333	333		Depreciation		333	333	333
23	EBIT		167	167	167		EBIT		167	167	167
24	Interest expense		32	22	11		Interest expense		0	0	0
25	EBT		135	145	155		EBT		167	167	167
26	Income tax		0	0	0		Income tax		0	0	0
27	Interest tax shield		0	0	0		Interest tax shield		0	0	0

Financial discount rates: No income tax

	A	B	C	D	E	F	G	H	I	J	K
30	Year	0	1	2	3		Year	0	1	2	3
31	Cash receipts		1000	1000	1000		Cash receipts		1000	1000	1000
32	Revenues		1000	1000	1000		Revenues		1000	1000	1000
33	Cash expenditures	1000	500	500	500		Cash expenditures	1000	500	500	500
34	Investment cost	1000					Investment cost	1000			
35	Operating costs		500	500	500		Operating costs		500	500	500
36	Market net cash flow	-1000	500	500	500		Market net cash flow	-1000	500	500	500
37	Income tax		0	0	0		Income tax		0	0	0
38	Net cash flow (TIPV)	-1000	500	500	500		Net cash flow (AEPV)	-1000	500	500	500
39	Debt finance flow	400	-155	-155	-155		Debt at the beginning		400	277	144
40	Net cash flow (EPV)	-600	345	345	345		Interest charge		32	22	11
41	Value of project (V^L)	1053	764	417			Debt repayment		155	155	155
42	Value of debt (D)	400	277	144			Debt at the end	400	277	144	0
43	Value of equity (E^L)	653	487	273			Discount factor, EPV	1	0.79	0.62	0.49
44	Debt-Equity ratio (D/E)	0.61	0.57	0.53			Discount factor, TIPV-AEPV	1	0.83	0.69	0.58
45	%D	38.0%	36.2%	34.5%			PV[NCF-TIPV]@ $wacc_1 = \rho = 20\%$	-1000	416.7	347.2	289.4
46	%E	62.0%	63.8%	65.5%			PV[NCF-EPV]@ e	-600	270.7	213.5	169.0
47	e	27.3%	26.8%	26.3%			Points of view	TIPV	AEPV	EPV	Debt
48	$wacc_1 = \rho$	20.0%	20.0%	20.0%			NPV	53.2	53.2	53.2	0.0

Financial discount rates

Impact of income tax



Financial discount rates: Income tax

- With taxes, some adjustments have to be made in the above formulas.
- The value of the levered project is now equal to the value of the unlevered project plus the present value of the tax shield.
- In particular,

$$V^L = V^{UL} + PV(TS) = E^L + D$$

- It is commonly assumed that the appropriate discount rate for the tax shield is $wacc_1$, cost of capital.

Financial discount rates: Income tax

- The value of WACC that is used for discounting the total investment cash flow will depend on whether the tax shield is or excluded or included.
- If the tax shield is excluded, then in the construction of the income statement, the interest deduction will be excluded in order to determine the tax liability as if there was no financing.
- If the tax shield is included, then in the construction of the income statement, the interest deduction will be included in order to determine the correct tax liability.

Financial discount rates: Income tax

Method 1: Including the tax shield and using NCF_{TIPV}

In this case, the cash flow statement includes the value of the tax shield. Consequently, there is no need to adjust the expressions for the calculation of the WACC and the return to equity e .

$$wacc_1 = \%Dd + \%Ee = \rho_1$$

$$e_1 = \rho_1 + (\rho_1 - d) \frac{D}{E}$$

Financial discount rates: Income tax

Method 2: Excluding the tax shield and using $NCF_{AE PV}$

Since the cash flow statement does not include the tax shield, the value of the tax shield is taken into account in the WACC by multiplying the cost of debt d by the factor $(1 - t)$, t is the income tax rate.

$$wacc_2 = \%Dd(1 - t) + \%Ee = \rho_2$$

$$e_2 = \rho_2 + (\rho_2 - d)\frac{D}{E}$$

Financial discount rates: Income tax

- $NCF_{TIPV} = NCF_{AEPV} + TS$
- $V_{wacc1}^L = V_{wacc1}^{UL} + PV_{wacc1}^{TS}$
- $NPV_{TIPV}^{wacc1} = NPV_{AEPV}^{wacc1} + PV_{TS}^{wacc1} = NPV_{AEPV}^{wacc2}$
- $NPV_{TIPV}^{wacc1} = NPV_{EPV}^{e1} = NPV_{EPV}^{e2}$

So, we have:

$$NPV_{TIPV}^{wacc1} = NPV_{AEPV}^{wacc2} = NPV_{AEPV}^{wacc1} + PV_{TS}^{wacc1} = NPV_{EPV}^{e1} = NPV_{EPV}^{e2}$$

$$NCF_{EPV} = NCF_{AEPV} + \text{Debt finance flow}$$

$$NCF_{EPV} = NCF_{TIPV} + \text{Debt finance flow}$$

Financial discount rates: Income tax

- **Remarks:**

- $wacc_1 = \rho$ (constant during the project's life).
- $wacc_2 = \%Dd(1-t) + \%Ee = wacc_1 - ts$ ($ts = \%Ddt$; $wacc_2$ increases as $D/\bar{E}$ decreases) => not constant.
- $e [= \rho + (\rho - d)D/\bar{E}]$ decreases as $D/\bar{E}$ decreases => not constant.

$$NPV_{EPV} = \sum_{t=0}^n \frac{NCF_t^{EPV}}{\prod_t (1 + e_t)}$$

$$= NCF_0^{EPV} + \frac{NCF_1^{EPV}}{(1 + e_1)} + \frac{NCF_2^{EPV}}{(1 + e_1)(1 + e_2)} + \dots + \frac{NCF_n^{EPV}}{(1 + e_1)(1 + e_2) \dots (1 + e_n)}$$

Financial discount rates: Income tax

Year	0	1	2	3	4	5	6	7	8	9
NCF_{TIPV}	NCF_0	NCF_1	NCF_2	NCF_3	NCF_4	NCF_5	NCF_6	NCF_7	NCF_8	NCF_9
V^L	$=NPV(wacc_1, NCF_1:\$NCF_9)$									
D	$=$ The ending debt flow (loan schedule).									
E^L	$=V^L - D$									
D/E	$=D/E$									
e	$=\rho + (\rho - d)*D/E$, where ρ and d are given.									
$wacc_1$	$=D/V^L*d + E/V^L*e$									
DF_{EPV}	$DF_0 = 1$	$=DF_0/(1 + e_0)$	$=DF_1/(1 + e_1)$	...					$=DF_8/(1 + e_8)$	
PV_{EPV}	$=NCF_t^{EPV}*DF_t^{EPV}$									

Use
SUMPRODUCT(,)
to calculate NPV

Use
SUMPRODUCT(,)
to calculate NPV

Financial discount rates: Income tax

Year	0	1	2	3	4	5	6	7	8	9
NCF_{TIPV}	NCF_0	NCF_1	NCF_2	NCF_3	NCF_4	NCF_5	NCF_6	NCF_7	NCF_8	NCF_9
V^L	$=NPV(wacc_1, NCF_1:\$NCF_9)$									
D	$=$ The ending debt flow (loan schedule)									
E^L	$=V^L - D$									
D/E	$=D/E$									
e	$=\rho + (\rho - d)*D/E$, where ρ and d are given									
wacc ²	$=D/V^L*d*(1-t) + E/V^L*e$ (Note: wacc ² means wacc ₂)									
DF_{AEPV}	$DF_0 = 1$	$=DF_0/(1 + wacc^2_0)$	$=DF_1/(1 + wacc^2_1)$	...						$=DF_8/(1 + wacc^2_8)$
PV_{AEPV}	$=NCF_t^{AEPV}*DF_t^{AEPV}$									

Use
SUMPRODUCT(,)
to calculate NPV

Financial discount rates: Income tax

	A	B	C	D	E	F	G	H	I	J	K
5	TOTAL INVESTMENT POINT OF VIEW (TIPV)						ALL EQUITY POINT OF VIEW (AEPV)				
6	Revenues	1000					Revenues		1000		
7	Operating costs	500					Operating costs		500		
8	Depreciation	333					Depreciation		333		
12	Investment costs	1000	ρ 15%				Investment costs		1000		
13	Debt financing	60%					Debt financing		0%		
14	Interest rate	10%	Income tax rate 20%				Interest rate				
15	Interest expense	IPMT					Interest expense		0		
16											
17	<u>Income statement</u>						<u>Income statement</u>				
18	Year	0	1	2	3		Year	0	1	2	3
19	Revenues		1000	1000	1000		Revenues		1000	1000	1000
20	Operating costs		500	500	500		Operating costs		500	500	500
21	EBITDA		500	500	500		EBITDA		500	500	500
22	Depreciation		333	333	333		Depreciation		333	333	333
23	EBIT		167	167	167		EBIT		167	167	167
24	Interest expense		60	42	22		Interest expense		0	0	0
25	EBT		107	125	145		EBT		167	167	167
26	Income tax		21	25	29		Income tax		33	33	33
27	Interest tax shield		12	8	4		Interest tax shield		0	0	0

Financial discount rates: Income tax

	A	B	C	D	E	F	G	H	I	J	K
29	Cash flow statement						Cash flow statement				
30	Year	0	1	2	3		Year	0	1	2	3
31	<u>Cash receipts</u>		1000	1000	1000		<u>Cash receipts</u>		1000	1000	1000
32	Revenues		1000	1000	1000		Revenues		1000	1000	1000
33	<u>Cash expenditures</u>	1000	500	500	500		<u>Cash expenditures</u>	1000	500	500	500
34	Investment cost	1000					Investment cost	1000			
35	Operating costs		500	500	500		Operating costs		500	500	500
36	Market net cash flow	-1000	500	500	500		Market net cash flow	-1000	500	500	500
37	Income tax		21	25	29		Income tax		33	33	33
38	Net cash flow (TIPV)	-1000	479	475	471		Net cash flow (AEPV)	-1000	467	467	467
39	Debt finance flow	600	-241	-241	-241		Debt finance flow	600	-241	-241	-241
40	Net cash flow (EPV₁)	-400	237	234	230		Net cash flow (EPV₂)	-400	225	225	225
41	Value of project (V ^L)	1085	769	410			Debt at the beginning		600	419	219
42	Value of debt (D)	600	419	219			Interest charge		60	42	22
43	Value of equity (E ^L)	485	351	190			Debt repayment		241	241	241
44	Debt-Equity ratio (D/E)	1.24	1.19	1.15			Debt at the end	600	419	219	0
45	%D	55.3%	54.4%	53.5%			wacc ₂ = ρ ₂	13.9%	13.9%	13.9%	
46	%E	44.7%	45.6%	46.5%			Discount factor, EPV@e ₁	1	0.83	0.68	0.56
47	e ₁	21.2%	21.0%	20.8%			Discount factor, EPV@e ₂	1	0.84	0.71	0.60
48	e ₂	18.7%	18.6%	18.5%			Discount factor, TIPV@wacc1	1	0.87	0.76	0.66
49	wacc ₁ = ρ ₁	15.0%	15.0%	15.0%			Discount factor, AEPV@wacc2	1	0.88	0.77	0.68
50	Points of view	TIPV_{wacc1}	AEPV_{wacc2}	EPV_{e1}¹	EPV_{e2}²		Tax shield_{wacc1}	AEPV_{wacc1}	NPV^{AEPV}_{wacc1 + TS_{wacc1}}		
51	NPV	85.16	85.16	85.16	85.16		19.65	65.51	85.16		

Financial discount rates

Practical implications



Practical implications

- If a project is profitable from the viewpoint of a banker but unprofitable to the owner, the project could face problems during implementation.
- If there is inequality in the present values, what is the source of the inequality?
- In fact, a project has an unique value of NPV whatever it is appraised from the point of view of a banker or the owner. The sources of its divergence are the selection of discount rate and the impact of tax shield.

Practical implications

- Given the project data, the fact that the present values of two points of view are identical leads to the **first** implication:
 - ❖ Because the cost of equity e is time-variant, it's not convenient to use the NPV function.
 - ❖ Because the cost of capital $wacc_1$ is time-invariant, it's comfortable to estimate the present values with the below NPV function:

$$= NPV(wacc_1, NCF_1:NCF_n) + NCF_0$$

Practical implications

- The **second** implication from our analysis is the impact of income tax on the project values, and then the project net present values. This might provide the equity holders useful information about the way a project should be financed.
- The third implication from our analysis is that the analyst might check the internal consistency through comparing the present values of different points of view. If the present values are identical, the analyst can be fairly sure the analysis is correct, or at least, is internally consistent.

An illustrative example

NFG Project (Campbell & Brown, 2016, chapter 4)



	A	B	C	D	E	F	G	L	M	N	O	P	Q	R	S	T	U	V
61	CASH FLOW STATEMENT (TIPV)																	
62	Year	0	1	2	3	4	5	10	11	12	13	14	15	16	17	18	19	20
63	Receipts	0.0	81.3	162.5	243.8	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	399.0
64	Revenues		81.3	162.5	243.8	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0
65	Liquidation value																	74.0
66	Expenditures	740.0	136.7	100.6	150.9	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	114.9
67	Fixed investment cost	740.0																
68	Working capital investments	0	86	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-86
69	Operating costs	0.0	50.3	100.6	150.9	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2	201.2
70	Market net cash flow	-740.0	-55.4	61.9	92.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	284.1
71	Business income taxes	0.0	-16.0	-8.3	0.0	8.3	8.9	17.1	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8
72	Market net cash flow (TIPV)	-740.0	-39.4	70.1	92.8	115.5	114.9	106.7	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
73																		
74	CASH FLOW STATEMENT (EPV - NFG COMPANY)																	
75	Year	0	1	2	3	4	5	10	11	12	13	14	15	16	17	18	19	20
76	Market net cash flow (TIPV)	-740.0	-39.4	70.1	92.8	115.5	114.9	106.7	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
77	Debt finance flow	700.0	-44.2	-86.2	-86.2	-86.2	-126.2	-84.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
78	Private net cash flow (EPV)	-40.0	-83.6	-16.0	6.6	29.3	-11.3	22.5	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
79																		
80	OTHER CASH FLOWS																	
81	Year	0	1	2	3	4	5	10	11	12	13	14	15	16	17	18	19	20
82	Lender's net cash flow	-700.0	44.2	86.2	86.2	86.2	126.2	84.2										
83	Interest tax shield	0.0	6.1	6.1	5.6	5.0	4.4	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
84	Market net cash flow (AEPV)	-740.0	-45.5	64.0	87.3	110.5	110.5	106.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
85																		
94	CAPITAL STRUCTURE		Opp. cost	Weight														
95	Agricultural loan	700,000	3.5%	0.90														
96	Bank overdraft	40,000	5.0%	0.05														
97	Equity capital (NFG company)	40,000	10%	0.05														
98		780,000	3.9%															
99	wacc ₁		3.9%															

	A	B	C	D	E	F	G	L	M	P	Q	U	V
101	FINANCIAL DISCOUNT RATES												
102	Year	0	1	2	3	4	5	10	11	14	15	19	20
103	Net cash flow (TIPV)	-740.0	-39.4	70.1	92.8	115.5	114.9	106.7	96.0	96.0	96.0	96.0	256.3
104	Levered project values	1309.0	1399.6	1384.1	1345.5	1282.6	1217.9	891.1	830.0	631.9	560.7	246.7	
105	Value of debt 1 (agricultural loan)	700	640	579	515	448	380	0	0	0	0	0	
106	Value of debt 2 (bank overdraft)	0	40	40	40	40	0	0	0	0	0	0	
107	Levered value of equity capital	609.0	719.2	765.6	790.8	794.1	837.8	891.1	830.0	631.9	560.7	246.7	
108	Debt-Equity ratio 1	1.1	0.9	0.8	0.7	0.6	0.5	0.0	0.0	0.0	0.0	0.0	
109	Debt-Equity ratio 2	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	
110	Percentage of debt 1 (%D ₁)	0.53	0.46	0.42	0.38	0.35	0.31	0.00	0.00	0.00	0.00	0.00	
111	Percentage of debt 2 (%D ₂)	0.00	0.03	0.03	0.03	0.03	0.00	0.00	0.00	0.00	0.00	0.00	
112	Percentage of equity capital (%E)	0.47	0.51	0.55	0.59	0.62	0.69	1.00	1.00	1.00	1.00	1.00	
113	Cost of equity (e)	4.4%	4.2%	4.2%	4.1%	4.1%	4.1%	3.9%	3.9%	3.9%	3.9%	3.9%	
114	Cost of capital (wacc ₁)	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	3.9%	
115	Cost of capital (wacc ₂)	3.4%	3.5%	3.5%	3.5%	3.6%	3.6%	3.9%	3.9%	3.9%	3.9%	3.9%	
116	Discount factor _{@e}	1	0.96	0.92	0.88	0.85	0.81	0.67	0.64	0.57	0.55	0.47	0.46
117	Discount factor _{@wacc1}	1	0.96	0.93	0.89	0.86	0.83	0.68	0.66	0.58	0.56	0.48	0.46
118	Discount factor _{@wacc2}	1	0.97	0.93	0.90	0.87	0.84	0.70	0.67	0.60	0.58	0.50	0.48
119													
120	Financial points of view	TIPV	AEPV	EPV	TS	AEPV							
121	Financial discount rates	wacc ₁	wacc ₂	e	wacc ₁	wacc ₁							
122	Net present values	569.0	569.0	569.0	32.2	536.8							
123	Internal rates of return	10.1%	9.7%	19.1%									

Financial discount rates

Q&A section

